29 - (MathAA-W 2021-Paper 2/0-Q10) **-** *ALGEBRA-FUNCTIONS - ROOTS, GRAPHS, INTEGRATION*

[Maximum mark: 18]

Consider the function $f(x)=\dfrac{x^2-x-12}{2x-15}$, $x\in\mathbb{R}$, $x\neq\dfrac{15}{2}$.

(a) Find the coordinates where the graph of f crosses the

(i) x-axis;

(ii) y-axis. [3]

(b) Write down the equation of the vertical asymptote of the graph of f. [1]

(c) The oblique asymptote of the graph of f can be written as $y=ax+b$ where a, $b\in\mathbb{Q}$.

Find the value of a and the value of b. [4]

(d) Sketch the graph of f for $-30\le x\le 30$, clearly indicating the points of intersection with each axis and any asymptotes. [3]

(e) (i) Express $\dfrac{1}{f(x)}$ in partial fractions.

(ii) Hence find the exact value of $\displaystyle\int_0^3 \frac{1}{f(x)}\,\mathrm{d}x$, expressing your answer as a single logarithm. [7]